MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **PERIODIC ASSESSMENT TEST – 08 (LINEAR EQUATIONS)** **Time:** 60 Minutes

Class: Class X (CBSE) **Chapter 3: Pair of Linear Equations in Two Variables** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 14(iv).
2. **Sec A:** 4 MCQs (1M). **Sec B:** 2 AR (1M). **Sec C:** 3 VSAQ (2M). **Sec D:** 3 SA (3M). **Sec E:** 1 LA (5M). **Sec F:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (4 × 1 = 4 Marks)

Q1. If the system $kx + 3y = k - 3$ and $12x + ky = k$ has infinitely many solutions, then k is: [1]
(A) 6 (B) -6 (C) 0 (D) 12

Q2. There are 20 cars and motorcycles in a parking area. If there are 56 wheels altogether, how many cars are there? [1]
(A) 12 (B) 10 (C) 8 (D) 14

Q3. The sum of two numbers is 20 and their product is 75. The sum of their reciprocals is: [1]
(A) 1/15 (B) 4/15 (C) 3/10 (D) 2/5

Q4. The lines represented by $x - 4y + 14 = 0$ and $3x + 2y - 14 = 0$ intersect at the point: [1]
(A) $(2, 4)$ (B) $(-2, 4)$ (C) $(4, 2)$ (D) $(2, -4)$

SECTION B: Assertion-Reason Questions (2 × 1 = 2 Marks)

Directions: (A) Both (A) and (R) are true, and (R) is correct explanation. (B) Both true, but (R) is not correct explanation. (C) (A) is true, (R) is false. (D) (A) is false, (R) is true.

Q5. **Assertion (A):** The pair of equations $x+2y-5 = 0$ and $2x+4y-10 = 0$ has infinitely many solutions.
Reason (R): If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, then the linear equations represent coincident lines. [1]

Q6. **Assertion (A):** The point $(0, 4)$ lies on the straight line $3x + 2y = 8$.
Reason (R): Substituting $x = 0$ and $y = 4$ into $3x + 2y$ yields $3(0) + 2(4) = 8$. [1]

SECTION C: Very Short Answer Questions (3 × 2 = 6 Marks)

Q7. Find the value of k for which the lines $5x - 7y = 3$ and $15x + ky = 9$ represent coincident lines. [2]

Q8. Solve for x and y: $0.2x + 0.3y = 1.3$ and $0.4x + 0.5y = 2.3$. [2]

Q9. For what value of p does the pair $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a unique solution? [2]

SECTION D: Short Answer Questions (3 × 3 = 9 Marks)

Q10. Solve for x and y: $a(x+y)+b(x-y)=a^2-ab+b^2$ and $a(x+y)-b(x-y)=a^2+ab+b^2$. [3]

Q11. Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. What are the speeds of the two cars? [3]

Q12. The area of a rectangle gets reduced by 9 sq. units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 sq. units. Find the dimensions of the rectangle. [3]

SECTION E: Long Answer / HOTS Question (1 × 5 = 5 Marks)

Q13. The sum of a two-digit number and the number formed by reversing the digits is 132. If 12 is added to the number, the new number becomes 5 times the sum of the digits. Find the original number. [5]

SECTION F: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q14. Case Study: Multilevel Automated Parking Vehicle Toll Matrix

At a modern multilevel automated parking complex near a metro station, electronic sensors record vehicles entering the basement. The complex accommodates two-wheelers (scooters/motorcycles) and four-wheelers (cars).

During the morning peak hour, the entrance boom barrier registered 150 vehicles entering the lot. The underground pressure sensors recorded a total of 420 rolling wheels. Let x denote the number of two-wheelers and y denote the number of four-wheelers.

Based on the above information, answer the following questions:

(i) Formulate the pair of linear equations in x and y modeling the vehicle count and wheel count. [1]

(ii) Determine the total number of two-wheelers parked in the facility (x). [1]

(iii) Determine the total number of four-wheelers parked in the facility (y). [1]

(iv) If the parking fee is Rs. 20 per two-wheeler and Rs. 50 per car, calculate the total revenue generated during the morning rush. [1]

OR

Find the difference between the revenue collected from cars and two-wheelers. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★